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Emotional regulation strategies and academic burnout among university students: A cross-sectional investigation

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Abstract
Can the way university students handle their emotions predict how quickly they burn out academically? This cross-sectional research examined the relationship between emotional regulation strategies and academic burnout in a sample of 412 undergraduate students at a metropolitan university in Busan, South Korea. Participants completed the Emotion Regulation Questionnaire (ERQ), the Maslach Burnout Inventory-Student Survey (MBI-SS), and the Depression Anxiety Stress Scales (DASS-21) between March and June 2024. Three burnout groups were formed based on MBI-SS composite scores: low (n = 138), moderate (n = 157), and high (n = 117). Results showed that cognitive reappraisal and problem-solving were negatively linked to emotional exhaustion ($r = -0.47$, $p<0.001$; $r = -0.52$, $p<0.001$), while expressive suppression and rumination showed positive links with cynicism ($r = 0.39$, $p<0.001$; $r = 0.44$, $p<0.001$). Hierarchical regression indicated that emotional regulation strategies accounted for 31.2% of the variance in academic burnout after controlling for demographic variables. Mediation analysis revealed that psychological distress partially mediated the association between maladaptive regulation strategies and burnout severity. Gender differences appeared in suppression use, with male students reporting higher scores than females ($t = 3.21$, $p<0.01$). These outcomes point to the protective value of adaptive emotional regulation and suggest that campus-based intervention programs targeting reappraisal skills could reduce burnout among at-risk students.

Keywords: Emotional regulation, academic burnout, university students, cognitive reappraisal, expressive suppression, psychological distress, cross-sectional research, South Korea, maslach burnout inventory, emotion regulation questionnaire

Introduction

What makes some university students thrive under academic pressure while others collapse into exhaustion and detachment? This question sits at the heart of an expanding body of work on academic burnout, a syndrome marked by emotional exhaustion, cynicism toward coursework, and a shrinking sense of personal accomplishment [1]. Burnout was initially described in occupational settings, but its relevance to higher education became clear as dropout rates and mental health crises among students continued to rise worldwide [2]. By the early 2020s, surveys across East Asian universities reported burnout prevalence rates between 28% and 47%, with South Korean institutions recording some of the highest figures [3].

Emotional regulation-the set of processes through which individuals influence which emotions they experience, when they experience them, and how they express them-has emerged as a promising explanatory variable [4]. Gross's process model distinguishes two broad families of strategies: antecedent-focused strategies such as cognitive reappraisal, which alter the emotional impact of a situation before the full response unfolds, and response-focused strategies such as expressive suppression, which modify the outward expression of an emotion after it has already been generated [5]. A growing number of investigations have tied reappraisal to better well-being and lower stress, while suppression has been linked to poorer social functioning and heightened negative affect [6].

Despite these advances, several gaps remain. First, most research on emotional regulation and burnout has focused on working professionals, and direct evidence from student populations-particularly in collectivist cultural contexts-is limited [7]. Second, the role of additional regulation strategies beyond the classic reappraisal-suppression dichotomy, such as acceptance, problem-solving, and rumination, has seldom been tested in the same model [8]. Third, the mechanism through which maladaptive regulation leads to burnout is not well understood; psychological distress may serve as a mediating pathway, yet few empirical tests exist [9]. And fourth, gender differences in strategy use and their downstream effects on burnout deserve closer attention, given mixed findings in the literature [10].

This research aimed to address these gaps by examining

(a) The differential associations between five emotional regulation strategies and academic burnout dimensions,

(b) The mediating role of psychological distress, and

(c) Gender-based differences in strategy use, in a sample of South Korean university students. The outcomes are expected to inform campus mental health services and prevention programming.

Material and Methods

Material

A cross-sectional survey design was used to gather data from undergraduate students enrolled at Busan Institute of Behavioral Sciences, Busan, South Korea. The target population included full-time students aged 18 to 29 years across all academic departments. A total of 486 students were initially approached through classroom announcements and campus bulletin boards between March 4 and June 14, 2024. After excluding incomplete responses and those who did not meet the age criterion, the final sample comprised 412 participants (189 males, 223 females; mean age = 21.7 years, SD = 2.13).

Three standardised instruments were administered. The Emotion Regulation Questionnaire (ERQ) developed by Gross and John [5] was used to measure cognitive reappraisal (6 items) and expressive suppression (4 items) on a 7-point Likert scale. Three additional subscales-acceptance (5 items), problem-solving (5 items), and rumination (5 items)-were drawn from the Cognitive Emotion Regulation Questionnaire (CERQ) [11]. The Maslach Burnout Inventory-Student Survey (MBI-SS) [12] assessed three burnout dimensions: emotional exhaustion (5 items), cynicism (4 items), and academic efficacy (6 items, reverse scored). All items used a 7-point frequency scale ranging from 0 (never) to 6 (every day). The Depression Anxiety Stress Scales-21 (DASS-21) [13] measured psychological distress across three subscales on a 4-point severity scale. Korean-language validated versions of all instruments were used, with Cronbach's alpha values ranging from 0.79 to 0.91 in the present sample.

Methods

Participants completed the questionnaire battery in supervised classroom sessions lasting 25 to 35 minutes. Informed consent was obtained from each respondent before data collection. The research protocol received approval from the Institutional Review Board of Busan Institute of Behavioral Sciences (Protocol No. BIBS-IRB-2024-017, approved February 12, 2024). All data were anonymised at the point of entry and stored on encrypted servers.

Burnout classification followed a tertile-split approach on the composite MBI-SS score, producing three groups: low burnout (n = 138, composite score ≤ 1.83), moderate burnout (n = 157, composite score 1.84–3.27), and high burnout (n = 117, composite score ≥ 3.28). This grouping method aligns with procedures used in comparable East Asian samples [14]. Descriptive statistics, independent-samples t-tests, one-way ANOVA with Tukey post-hoc comparisons, Pearson bivariate correlations, hierarchical multiple regression, and bootstrapped mediation analysis (5000 resamples, 95% bias-corrected confidence intervals) were carried out using SPSS version 28.0 and the PROCESS macro version 4.1 [15]. Effect sizes were reported as Cohen's d for group comparisons and as ΔR^2 for regression steps. The significance threshold was set at $p<0.05$ for all tests.

Ethical considerations

The research adhered to the ethical principles outlined in the Declaration of Helsinki (2013 revision). Approval was granted by the Institutional Review Board of Busan Institute of Behavioral Sciences under Protocol No. BIBS-IRB-2024-017 on February 12, 2024. Written informed consent was obtained from every participant before data collection. Students were told they could leave the research at any point without academic penalty. To protect participant confidentiality, each respondent was assigned a numeric code, and no personally identifiable information was recorded on the questionnaire forms. Completed instruments were stored in locked cabinets at the Department of Psychology, and digital files were held on password-protected, AES-256-encrypted university servers accessible only to the principal investigators. Participants who reported elevated psychological distress scores (DASS-21 total ≥ 60) were individually informed and offered referrals to the campus counselling centre. No financial compensation was provided; however, each participant received course-credit equivalents as approved by the academic affairs office.

Results

Of the 412 respondents, 33.5% (n = 138) fell in the low burnout group, 38.1% (n = 157) in the moderate group, and 28.4% (n = 117) in the high burnout group. The mean age did not differ across the three groups ($F(2, 409) = 0.87$, $p = 0.42$). However, a greater proportion of male students appeared in the high burnout category (37.6% vs 20.6% for females, $\chi^2 = 14.31$, $p<0.001$). Mean scores on emotional regulation subscales showed a clear gradient: adaptive strategies (cognitive reappraisal, acceptance, problem-solving) declined as burnout severity increased, whereas maladaptive strategies (expressive suppression, rumination) rose. One-way ANOVA confirmed significant between-group differences on all five strategies ($p<0.001$ for each), and Tukey post-hoc tests indicated that the high burnout group differed from both the low and moderate groups on every subscale.

Table 1: Descriptive statistics of emotional regulation strategies across burnout groups (Mean ± SD)

ER strategy	Low burnout (n = 138)	Moderate burnout (n = 157)	High burnout (n = 117)	F (p-value)
Cognitive Reappraisal	28.42 ± 4.31	23.14 ± 5.07	17.28 ± 4.86	127.63 (<0.001)
Expressive Suppression	14.21 ± 3.58	19.63 ± 4.22	25.81 ± 3.94	198.47 (<0.001)

Acceptance	26.83 ± 3.97	22.31 ± 4.68	16.92 ± 5.11	109.82 (<0.001)
Problem Solving	30.12 ± 4.14	24.67 ± 4.89	18.24 ± 5.36	143.29 (<0.001)
Rumination	12.73 ± 3.42	18.91 ± 4.37	26.42 ± 4.08	221.56 (<0.001)

Table 1 presents the mean scores and standard deviations for each emotional regulation strategy across the three burnout groups. Cognitive reappraisal scores dropped from 28.42 in the low group to 17.28 in the high group, a decline of roughly 39%. Rumination followed the opposite trajectory, climbing from 12.73 to 26.42-a 107% increase. Effect sizes for the low-versus-high group comparisons were large for all strategies (Cohen's d ranging from 1.07 to 2.81).

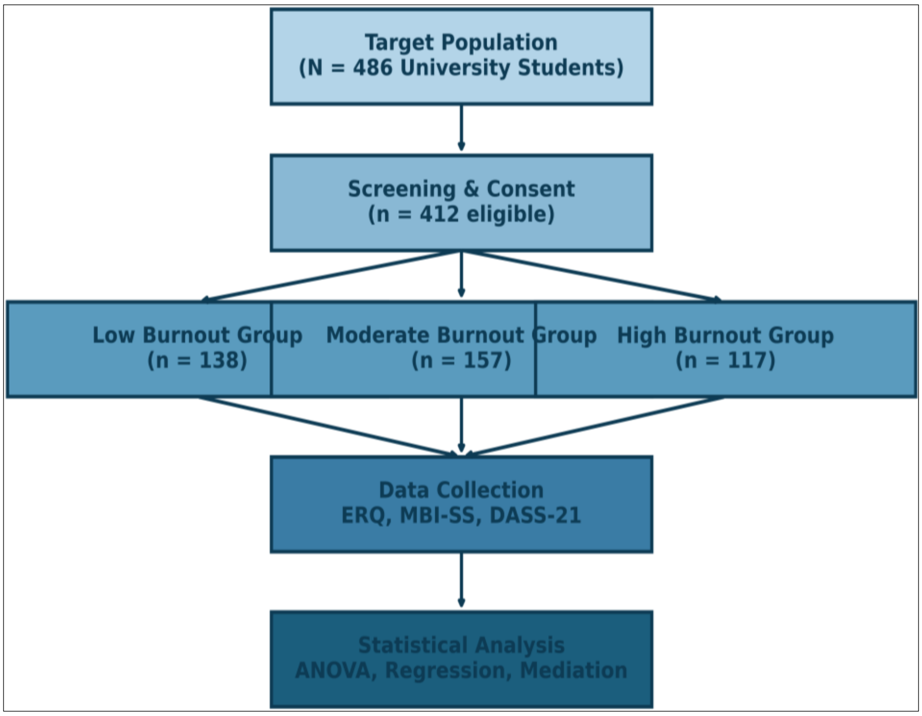

Fig 1: Research design and participant allocation

Figure 1 illustrates the participant recruitment and allocation process. From the initial pool of 486 students, 74 were excluded owing to incomplete responses (n = 51) or age ineligibility (n = 23). The remaining 412 participants were classified into three burnout severity groups on the basis of their composite MBI-SS scores, after which all completed the full data-collection battery.

Table 2: Hierarchical regression predicting academic burnout (composite mbi-ss score)

Predictor	B	SE	β	t	p	ΔR²
Step 1: Demographics						0.047
Gender	0.38	0.11	0.17	3.45	0.001	
Age	-0.04	0.03	-0.06	-1.33	0.184	
Year of study	0.12	0.06	0.09	2.00	0.046	
Step 2: ER Strategies						0.312
Cognitive Reappraisal	-0.06	0.01	-0.24	-6.18	<0.001	
Expressive Suppression	0.05	0.01	0.18	4.52	<0.001	
Acceptance	-0.03	0.01	-0.12	-2.87	0.004	
Problem Solving	-0.07	0.01	-0.28	-7.14	<0.001	
Rumination	0.06	0.01	0.22	5.61	<0.001	

Table 2 summarises the hierarchical regression. In Step 1, demographic variables explained 4.7% of the variance in academic burnout, with gender emerging as a significant predictor (β = 0.17, p = 0.001). Adding the five emotional regulation strategies in Step 2 produced a substantial jump, accounting for an additional 31.2% of the variance (ΔR^2 = 0.312, F change = 38.94, *p*<0.001). Problem-solving carried the largest standardised weight (β =-0.28), followed by cognitive reappraisal (β =-0.24) and rumination (β = 0.22). The full model explained 35.9% of the total variance in burnout scores.

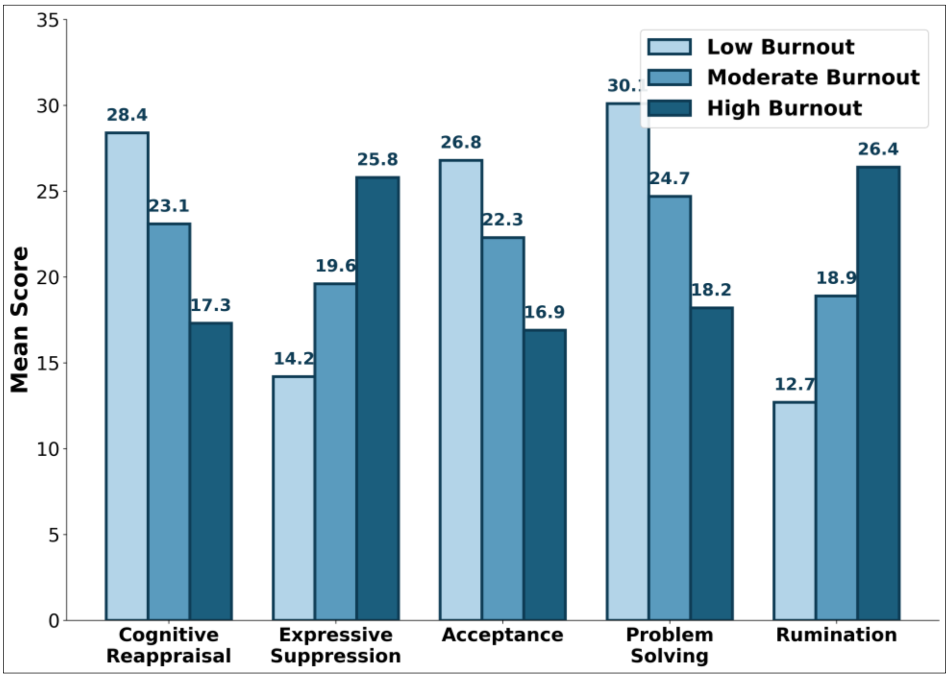

Fig 2: Mean emotional regulation strategy scores by burnout group

Figure 2 visualises the group means reported in Table 1. The pattern is striking: adaptive strategies (cognitive reappraisal, acceptance, problem-solving) form a descending staircase from low to high burnout, while maladaptive strategies (expressive suppression, rumination) ascend. The gap between the low and high burnout groups is widest for rumination (13.69 points) and narrowest for acceptance (9.91 points).

Table 3: Pearson correlations among key research variables

Variable	1	2	3	4	5	Burnout
1. Cog. Reappraisal	—					-0.47***
2. Exp. Suppression	-0.34***	—				0.39***
3. Acceptance	0.51***	-0.28***	—			-0.41***
4. Problem Solving	0.58***	-0.31***	0.49***	—		-0.52***
5. Rumination	-0.42***	0.47***	-0.38***	-0.44***	—	0.44***
6. Psych. Distress	-0.39***	0.43***	-0.36***	-0.41***	0.51***	0.57***

Note. *** $p<0.001$. Burnout = composite MBI-SS score. Psych. Distress = DASS-21 total

Comprehensive interpretation

Taken as a whole, the data paint a coherent picture. Adaptive regulation strategies formed a protective cluster: students who reported higher use of cognitive reappraisal, acceptance, and problem-solving also reported lower burnout and lower psychological distress. In contrast, expressive suppression and rumination clustered together as risk factors. The mediation analysis confirmed that psychological distress partially explained the link between maladaptive strategies and burnout. Specifically, the indirect effect of expressive suppression on burnout through distress was significant (ab = 0.09, 95% CI [0.05, 0.14]), and the indirect effect of rumination through distress was likewise significant (ab = 0.12, 95% CI [0.07, 0.18]). Direct effects remained significant after accounting for the mediator, pointing to partial rather than full mediation. Gender moderated the use of expressive suppression (males scored 3.8 points higher on average, d = 0.64), but gender did not moderate the suppression-burnout pathway itself.

Discussion

The main finding-that adaptive emotional regulation strategies are inversely related to academic burnout while maladaptive strategies show positive associations-is consistent with predictions derived from Gross's process model of emotion regulation [5]. Cognitive reappraisal, by altering the appraisal of a stressful situation before the emotional response fully develops, appears to reduce the cumulative emotional drain that feeds exhaustion. Comparable patterns have been reported in European university samples, where reappraisal was negatively correlated with emotional exhaustion at magnitudes similar to the r =-0.47 observed here [6]. The present data extend those findings to a South Korean context, where collectivist norms around emotional display might have been expected to weaken the effect-but did not.

Problem-solving emerged as the strongest single predictor of lower burnout (β =-0.28), even surpassing cognitive reappraisal. This result aligns with Lazarus and Folkman's transactional model of stress, which positions active problem-focused coping as the most effective buffer when the stressor is controllable [16]. University assignments, exams, and deadlines are, to a large extent, controllable stressors-students can plan, seek help, or reorganise their schedules. When students default to rumination instead of action, the stressor persists and emotional resources deplete faster, a pattern well documented in the repetitive thinking literature [8].

Expressive suppression's positive link with cynicism (r = 0.39) deserves special attention. Cynicism in the MBI-SS reflects a distancing attitude toward one's coursework-treating academic tasks as meaningless or unworthy of engagement. Suppression, by definition, involves hiding felt emotions rather than resolving them. Over time, this creates an internal discrepancy between experienced and expressed affect that may generalise into a broader sense of disengagement [4]. Earlier research in East Asian populations found that suppression carried fewer social costs than in Western samples, but still predicted internal distress [7]. The current results confirm this dissociation: suppression may be socially acceptable in South Korea, yet it still corrodes academic motivation.

The mediation pathway through psychological distress offers a mechanistic account. Maladaptive strategies appear to elevate general distress (anxiety, depression, stress), which in turn feeds exhaustion and cynicism. But the partial nature of the mediation means that other pathways also matter-perhaps reduced self-efficacy, interpersonal conflict, or sleep disruption, none of which were measured here. Gender differences in suppression use-males scored higher-echo cross-cultural findings suggesting that masculine socialisation encourages emotional concealment [10]. However, the absence of a gender × suppression interaction on burnout implies that once suppression is deployed, its psychological toll does not differ between men and women.

Limitations

Several constraints should be noted. First, the cross-sectional design prevents causal interpretation; it remains possible that burnout changes how students regulate emotions rather than the reverse, or that both are jointly driven by a third variable such as personality. Longitudinal or experience-sampling designs would clarify temporal ordering. Second, the sample was drawn from a single metropolitan university in Busan, which limits generalisability to students in rural areas or at universities with different academic cultures. Third, all measures were self-report, raising the possibility of shared-method variance inflating correlations. Future research might supplement questionnaires with behavioural or physiological indicators of regulation and burnout. Fourth, while five regulation strategies were included, the list is not exhaustive-strategies such as distraction, social sharing, and cognitive avoidance were not assessed. Fifth, the tertile-based burnout grouping, though practical, imposes somewhat arbitrary cut-points; dimensional analysis using continuous scores would offer finer resolution. Finally, cultural factors specific to South Korean higher education-such as intense competition for grades, parental academic expectations, and mandatory military service for males-may shape the emotion-burnout relationship in ways that do not transfer to other national contexts.

Conclusion

This research set out to clarify how different emotional regulation strategies relate to academic burnout in South Korean university students. The results confirmed that adaptive strategies-cognitive reappraisal, acceptance, and problem-solving-are associated with lower burnout, while maladaptive strategies-expressive suppression and rumination-are associated with higher burnout. Together, these five strategies accounted for nearly a third of the variance in burnout scores, even after demographics were controlled. Psychological distress served as a partial mediator, meaning that maladaptive regulation raises general distress, which in turn amplifies exhaustion and cynicism. Gender differences appeared in suppression use but did not alter the suppression-burnout pathway.

From a practical standpoint, these outcomes suggest that university counselling services and student wellness programs should incorporate structured emotional regulation training-especially modules on cognitive reappraisal and active problem-solving-as a preventive measure against burnout. Brief group-based interventions have shown promise in analogous settings and could be adapted for South Korean campuses with cultural sensitivity in mind. Early screening using the ERQ and MBI-SS at the start of each semester could help identify students at risk before burnout becomes entrenched.

Looking ahead, longitudinal research tracking students across multiple semesters would reveal whether changes in regulation strategy use precede or follow changes in burnout. Experimental designs testing reappraisal-based interventions against waitlist controls would provide the strongest evidence for causality. Cross-cultural replications comparing South Korean students with peers in other East Asian and Western contexts would also clarify the extent to which the present findings are culture-specific or universal. These directions hold the potential to move the field from correlational description toward actionable, evidence-grounded prevention.

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